Equivalent Expressions: Expanded and Factored Forms Answer Key

Grades 6–7 Expressions and Equations

Quick Answers

1. $3x + 12$	5. 27	9. 12
2. $5x + 15$	6. $12x + 18$	10. $8x - 20y$
3. $14 - 2x$	7. $-8x + 12y - 4$	11. $8x$
4. $7x + 6$	8. $6x + 6$	12. $18x + 24$, 60

Curriculum

Level: Grades 6–7 Expressions and Equations
6.EE.A / 7.EE.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

5. *If they got 19: distributed the 3 to the x only and left the 4 alone*
9. *If they got 27: multiplied only the first term inside the parentheses, writing 8x-5*
12. *If they got 180: divided only the first term by 6 and left the 24 inside the parentheses*

1. Expand $3(x + 4)$.

Step 1. The multiplier outside touches *every* term inside, so write two products before simplifying: $3 \cdot x$ and $3 \cdot 4$.
Step 2. $3 \cdot x = 3x$ and $3 \cdot 4 = 12$, so the sum is $3x + 12$. $\boxed{3x + 12}$
Why: $a(b + c) = ab + ac$ read left to right.

2. Factor $5x + 15$.

Step 1. Factoring is the same rule read backwards, so first find what both terms share: $5x = 5 \cdot x$ and $15 = 5 \cdot 3$, so the GCF is 5.
Step 2. Pull the 5 out front and write what is left inside: $5(x + 3)$. $\boxed{5(x + 3)}$
Check: expanding $5(x + 3)$ gives $5x + 15$ again.

3. Expand $-2(x - 7)$.

Step 1. Treat the subtraction as adding a negative, $x+(-7)$, so the multiplier -2 distributes over two terms and both signs are handled by the same rule.
Step 2. $-2 \cdot x = -2x$ and $-2 \cdot (-7) = +14$, so the result is $-2x + 14$. $\boxed{-2x + 14}$
The common error: writing $-2x - 14$. A negative times a negative is positive; the second sign flips.

4. Simplify $4x + 3x + 6$.
Step 1. Like terms must have the identical variable part, so sort the terms before adding anything: $4x$ and $3x$ are like; 6 has no x at all.
Step 2. $4x + 3x = 7x$, and the 6 is carried along unchanged: $7x + 6$. $\boxed{7x + 6}$
Why 6 cannot join: $7x$ and 6 count different things — $7x$ depends on x, 6 never does.

5. Ana's claim $3(x + 4) = 3x + 4$.
Step 1. A single substitution can disprove equivalence, so evaluate the correct expression first and use it as the reference value.
Step 2. At $x = 5$: $3(5 + 4) = 3 \cdot 9 = 27$. $\boxed{27}$
Ana's expression: $3 \cdot 5 + 4 = 19$. Because $27 \neq 19$, the two expressions are not equivalent. Her mistake was distributing the 3 to the x only and leaving the 4 untouched; the correct expansion is $3x + 12$.

6. Factor $12x + 18$ completely.
Step 1. "Completely" means the terms left inside must share no factor, so list the factors rather than grabbing the first common one: $12 = 2 \cdot 2 \cdot 3$ and $18 = 2 \cdot 3 \cdot 3$, giving a GCF of 6.
Step 2. $12x \div 6 = 2x$ and $18 \div 6 = 3$, so the factored form is $6(2x + 3)$. $\boxed{6(2x + 3)}$
Why $2(6x + 9)$ is not finished: $6x$ and 9 still share a factor of 3. Pulling that out as well returns $6(2x + 3)$.

7. Expand $-4(2x - 3y + 1)$.
Step 1. Three terms inside means three products, so write the inside as a sum of signed terms first: $2x + (-3y) + 1$.
Step 2. $-4 \cdot 2x = -8x$, $\quad -4 \cdot (-3y) = +12y$, $\quad -4 \cdot 1 = -4$, so the expansion is $-8x + 12y - 4$.
$\boxed{-8x + 12y - 4}$
Note: $-8x$ and $12y$ are not like terms, so nothing collects here.

8. Simplify $2(x + 3) + 4x$.
Step 1. Nothing can be collected while the parentheses stand, so expand before combining.
Step 2. $2(x + 3) = 2x + 6$, so the expression is $2x + 6 + 4x$.
Step 3. Collect the x-terms: $2x + 4x = 6x$, leaving $6x + 6$. $\boxed{6x + 6}$
Order matters: adding $2 + 4$ first and writing $6(x + 3)$ would be a different expression — it equals $6x + 18$.

9. Dev's claim $4(2x - 5) = 8x - 5$.
Step 1. Evaluate the correct form first so there is a reference value to compare against.
Step 2. At $x = 4$: $4(2 \cdot 4 - 5) = 4(8 - 5) = 4 \cdot 3 = 12$. $\boxed{12}$
Dev's expression: $8 \cdot 4 - 5 = 27$. One disagreement is enough: $12 \neq 27$, so the expressions are not equivalent. Dev multiplied only the first term inside; the correct expansion is $8x - 20$.

10. Factor $8x - 20y$ completely.
Step 1. The terms carry different letters, so no variable is common — only a numeric factor can come out. Compare $8 = 2 \cdot 2 \cdot 2$ with $20 = 2 \cdot 2 \cdot 5$: the GCF is 4.
Step 2. $8x \div 4 = 2x$ and $20y \div 4 = 5y$, and the subtraction sign stays inside: $4(2x - 5y)$.
$\boxed{4(2x - 5y)}$
Check: $4 \cdot 2x = 8x$ and $4 \cdot 5y = 20y$, so expanding returns the original.

11. Expand and simplify $-3(4-2x)+2(x+6)$.

Step 1. Two sets of parentheses, so distribute each one separately before any collecting; mixing the steps is where signs get lost.
Step 2. $-3(4-2x) = -12+6x$, and $2(x+6) = 2x+12$.
Step 3. Collect: the x-terms give $6x+2x = 8x$, and the constants give $-12+12 = 0$, so only $8x$ survives. $\boxed{8x}$
What is surprising and why: the constants are exact opposites, so the expression is a pure multiple of x — it is 0 when x is 0, no matter what.

12. Challenge: bundling $18x+24$ stickers.

Step 1. A bundle size is a common factor, and the largest bundle is the GCF, so factor rather than guess: $18 = 2\cdot 3\cdot 3$ and $24 = 2\cdot 2\cdot 2\cdot 3$ share $2\cdot 3 = 6$.
Step 2. $18x \div 6 = 3x$ and $24 \div 6 = 4$, so the factored form is $6(3x+4)$: six equal bundles of $3x+4$ stickers. $\boxed{6(3x+4)}$
Step 3. Substitute $x=2$ into the factored form: $6(3\cdot 2+4) = 6\cdot 10 = 60$. $\boxed{60}$
Part (c): the original gives $18\cdot 2+24 = 60$ as well, so the two agree here. That is strong evidence but not a proof — two different expressions can cross at one value. Equivalence means agreeing at *every* value, which is what the distributive property guarantees.
The trap: factoring 6 out of $18x$ only and leaving 24 inside gives $6(3x+24)$, which is 180 at $x=2$ — three times too many stickers.